

Module 18: Advanced Topics & Architecture

Duration: Ongoing | **Level:** Advanced

Service Layer Pattern

Keep controllers thin; business logic in services:

```
app/
|-- Services/
|   |-- TaskService.php
|   |-- DashboardService.php
|-- Repositories/           # Optional data access layer
|   |-- TaskRepository.php
|-- Actions/                # Single-purpose classes
|-- CompleteTask.php

class CompleteTask
{
    public function execute(Task $task, User $user): Task
    {
        $task->markComplete();
        event(new TaskCompleted($task, $user));
        return $task;
    }
}
```

Repository Pattern

Abstract database access:

```
interface TaskRepositoryInterface
{
    public function findOverdue(): Collection;
}

class EloquentTaskRepository implements TaskRepositoryInterface { ... }
```

Bind in AppServiceProvider:

```
$this->app->bind(TaskRepositoryInterface::class, EloquentTaskRepository::class);
```

Multi-Tenancy

For SaaS apps serving multiple organizations:

- [stanc/tenancy](#)
 - Database-per-tenant or shared database with tenant_id
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Microservices vs Monolith

Laravel excels as a **modular monolith**:

```
app/
|-- Domains/
|   |-- Tasks/
|   |   |-- Models/
|   |   |-- Controllers/
|   |   |-- Services/
|   |-- Billing/
```

Extract to microservices only when you have clear scaling boundaries.

Laravel Packages Ecosystem

Package	Purpose
spatie/laravel-permission	Roles & permissions
spatie/laravel-medialibrary	Media management
spatie/laravel-activitylog	Audit logging
laravel/scout	Full-text search (Algolia/Meilisearch)
laravel/cashier	Stripe subscriptions
laravel/socialite	OAuth login
barryvdh/laravel-debugbar	Dev debugging

Design Patterns in Laravel

Pattern	Laravel Feature
Factory	Model factories
Observer	Model observers
Strategy	Driver-based systems (cache, mail)
Decorator	Middleware
Command	Artisan commands, Jobs
Repository	Custom implementations

API-First & Mobile

1. Build API with Sanctum/Passport
 2. Use API Resources for consistent JSON
 3. Version your API (/api/v1/)
 4. Document with [Scramble](#) or OpenAPI
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Continuing Your Journey

Build Portfolio Projects

1. **Blog CMS** - posts, categories, comments, admin panel
2. **E-commerce** - products, cart, Stripe checkout
3. **SaaS Dashboard** - subscriptions, teams, billing
4. **Real-time Chat** - broadcasting, presence channels

Certifications & Community

- Laravel certification (official)
- Laracasts learning paths
- Laravel Live conferences
- Contribute to open source

Stay Current

- Read [Laravel News](#)

- Follow release notes each major version
 - Upgrade guide: laravel.com/docs/upgrade
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Congratulations!

You've completed the TaskForge Laravel curriculum from setup through production. You now have:

- 19 learning modules (00-18)
- A full-featured example application
- Automated test suite
- Deployment configurations
- Real-world patterns used in professional Laravel development

Keep building. Keep testing. Keep shipping.

Curriculum Complete

Return to [README](#) for the full roadmap and progress checklist.